

15150

DRAINAGE SPECIALITIES

15150**DRAINAGE SPECIALITIES****Contents**

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1.0 GENERAL

1.1 Reference

Conform to general requirements for Mechanical Services of Division fifteen.

1.2 Quality Assurance

1.2.1 Manufacturer qualification

Firms regularly engaged in manufacture of floor drains, cleanouts and roof drains of types and sizes required, whose products have been in satisfactory use in the similar service for not less than 5 years.

1.3 Submittals

1.3.1 Product data

Submit manufacturers technical product data, installation instructions and dimensioned drawings for each type of items.

1.4 Maintenance Data

Submit maintenance data as per manufacturer recommendations. Include this data, product data, and certification in maintenance manual.

2.0 PRODUCT

2.1 Materials

2.1.1 Floor Drains

- a) Floor traps: ABS/HDPE with rodding eye (110 dia top inlet, 75mm dia outlet) with minimum water seal of 7 cm.
- b) Floor drain: Cast iron body with heavy duty 150 x 150mm stainless steel gratings and covers. The open area of strainer shall be at least two third of the cross sectional area of the drain line to which it connects. Note wherever floor drain is indicated it shall be connected to floor trap unless other wise indicated.
- c) Funnel floor drain where specified shall include a nickel bronze funnel secured to the grating.
- d) Flower bed drain where specified shall be heavy duty, dome type drain coated cast iron body with bottom outlet complete with combination membrane flashing clamp/gravel guard, low silhouette dome and stainless steel mesh screen over dome, no floor trap is required.
- e) Floor trap with trap primer shall be normal floor drain as described above but with trap primer connection.

2.1.2 Roof Drains

Rain water outlets shall be epoxy coated cast iron body and dome, wide serrated flashing flanged, flashing clamp device with integral gravel stop. Rain water outlets shall be selected to suit the type of roof where installed and shall be provided with a no hub outlet for mechanical jointing to rain water pipe.

2.1.3 Clean Outs

- a) Floor type: Cast iron body with square or round adjustable, secured, nickel bronze top with 3mm tile recess.
- b) Wall type: Cast iron body with square nickel bronze smooth wall access cover and frame, 150 x 150mm.
- c) Clean out at high level shall be HDPE screw cap and washer.

2.1.4 Grease Interceptors

Large capacity Acid Resistant coated interior and exterior fabricated steel grease interceptor, with internal air relief by pass, bronze cleanout plug and visible double wall trap seal with removable pressure equalizing/flow diffusing baffle and sediment tray. Gasketed non skid secured cover with covered recessed lift handle and enzyme port having bronze plug, complete with flow control fitting. Furnished with inlet and outlet in high position. Size of grease interceptors shall be suitable for the volume of fixtures discharging into the interceptors.

2.1.5 Manholes/gully traps covers

All covers shall be ductile iron with black bitumen coating.

Manhole covers shall have clear opening of 600 x 600 mm.

All manholes and gully traps shall be vented as per drainage department requirements.

Gully trap covers shall have clear opening of 300 x 300 mm.

All manhole covers and frames shall comply with BS 497:1976 part 1.

a) Internal Manholes:

All internal manhole covers shall be bolt down air tight double seal recessed type, they shall be insitu concrete filled with surface finish to match surrounding floor. Covers shall be medium duty, locking and sealed.

b) External Manholes:

Covers in paved areas shall be medium duty, double seal, recessed type. They shall be insitu concrete/pavement filled with surface finish to match surrounding floor.

Covers subject to vehicular traffic shall be kite marked heavy duty, single seal, non-rock with square push fit seal plate complete with lifting key-holes.

2.2 Applicable Standards

Generally roof drains and floor drains shall be in compliance with ANSI standard A 112.21.2 and A 112.21.1 M respectively.

3.0 EXECUTION

3.1 General

- 3.1.1 All roof and floor drains shall be installed with water proof membrane extended up to and within the drain, and flashed to prevent any water penetration through the drain. Water proofing shall be carried out as recommended by the manufacturer and as approved by the Engineer.
- 3.1.2 Install roof drains at low points of roof areas according to roof membrane manufacturer's written installation instructions. Maintain integrity of water proof membrane where penetrated.
- 3.1.3 Position roof drains and floor drains for easy access and maintenance.
- 3.1.4 Unless otherwise mentioned provide traps at each drain to stop seepage of sewer gases into occupied spaces.
- 3.1.5 Provide transition piece with mechanical joint connectors to couple drains to pipes specified.
- 3.1.6 Install floor drains at low points of surfaces to be drains, set grates of drains flush with finish floor or as indicated.
- 3.1.7 Clean outs shall be installed at change of direction of drainage pipes inside building and where indicated on the drawings. Clean outs shall be not more than 15m apart in horizontal lines for pipe diameters of 100 mm and smaller, and 30 m apart for larger pipes.
- 3.1.8 A clean out shall be provided at the foot of each vertical waste or soil stack and at each floor, the clean out shall be at 0.6 m height from finished floor level.
- 3.1.9 Clean outs on concealed piping shall be extended through and terminate flush with finished wall or floor. Pits or chases may be left in walls or floors provided they are of sufficient size to permit removal of clean outs plug and proper cleaning of the system.
- 3.1.10 Where it is necessary to conceal a cleanout plug, a heavy duty stainless steel covering plate 200 x 200 mm shall be provided which permits ready access to the plug.
- 3.1.11 Cleanouts shall be of the same nominal size as the pipes up to 100 mm pipe diameters and not less than 100 mm for larger piping.
- 3.1.12 Cleanouts shall be so installed that there is a clearance of not less than 450 mm for the purpose of rodding and cleaning.
- 3.1.13 Protect cleanouts during construction period to avoid clogging with dirt and debris and avoid damages from traffics and construction works.

END OF SECTION